



Analysis of Crack Parameters and Electrical Property in Flexible Devices Under Bending State

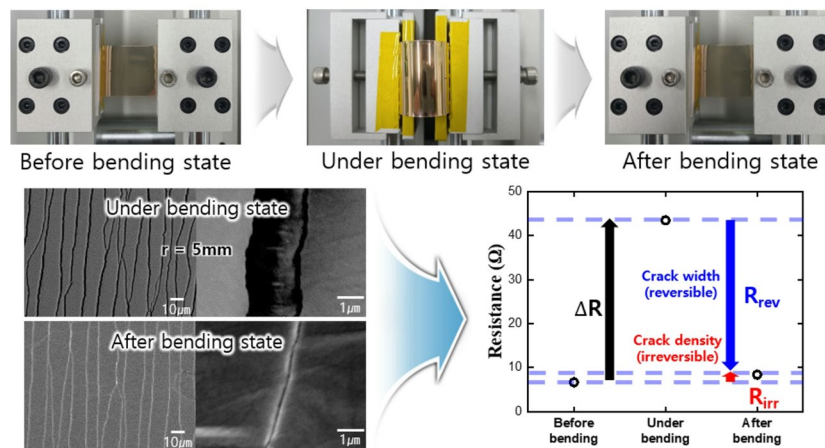
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Abstract

Repeated or sustained bending in flexible displays induces mechanical stress on all components, particularly affecting narrow and thin electronic circuits. There is a lack of research that maintains the bending state during observation of cracks and analyzes electrical properties based on crack parameters. This study aims to observe crack parameters under the bending state and analyze the relationship between crack parameters and electrical properties. The results showed that a smaller bending radius induced greater tensile stress on the specimen surface, which in turn increased both crack density and crack width. Measurement of resistance under bending state also revealed that smaller bending radius resulted in larger crack width and corresponding increases in resistance. We also demonstrated that analyzing cracks in the flat state (after bending state) significantly underestimated the effects of crack width in electrical properties because crack width was almost recovered indicating reversible deformation while crack density remained unchanged, representing irreversible damage after bending state. Therefore, crack analysis under actual bending state is essential for accurately evaluating the mechanical and electrical reliability of flexible electronic components.

Graphical Abstract



Keywords Bending · Crack width · Crack density · Resistance · Flexible device

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1 Introduction

The trend to adapt displays into curved or deformable forms is growing in various fields. Commercialized products such as rollable TVs and foldable smartphones exemplify this trend, offering the advantage of compact storage and large-area expansion during use. These types of displays are subjected to repeated bending deformation throughout their operation. On the other hand, curved displays used in automotive systems or architectural installations typically retain a fixed shape once installed, and experience sustained bending deformation over time during use. In both cases, whether they exhibit repeated or sustained bending deformation, not only the external form but also all internal components of the display are subjected to bending deformation. To ensure reliable functionality under such conditions, each internal component must possess adequate flexibility to withstand the imposed bending deformation.

Flexible displays typically consist of highly conductive metal for electrical interconnection and flexible polymer substrates for mechanical support. Mechanical bending deformation primarily induces damage, particularly cracking, in the electronic circuits rather than in the substrates. It is attributed to the circuits being relatively thin and less flexible compared to the thicker and more compliant polymer substrates. Many researchers investigated to understand the damage mechanisms of electronic circuits under bending deformation. The previous studies can be broadly categorized into repeated(or cyclic) bending and sustained bending. In repeated bending studies, researchers have analyzed variations in performance such as resistance [1–4], power efficiency [5–7], voltage [8, 9], and current [10]. In sustained bending studies, resistance [11–14], stress [15], surface strain [16], frequency [17], current [18, 19], and power efficiency [20] have been investigated. They mostly focused on performance changes derived from variations in electrical properties [1–14, 18–20]. These studies also have generally focused on identifying the causes of damage by measuring changes in electrical properties under repeated or sustained bending deformation. Many studies have observed cracks after bending state [2, 4–8, 10, 11], with only limited studies [3] directly analyzing cracks while maintaining the bending state due to methodological challenges. As a result, irreversible damage between under bending state and after bending state has been identified, however, reversible damage has not fully characterized. This limitation has restricted the quantitative analysis of electrical property changes using parameters directly related to cracks, which are the main cause of damage. In this study, we propose a novel method to observe cracks under bending state and investigate the correlation between crack parameters and electrical properties at various bending radii. It allows us to analyze how

variations in bending conditions affect both crack behavior and electrical performance.

2 Theoretical Analysis

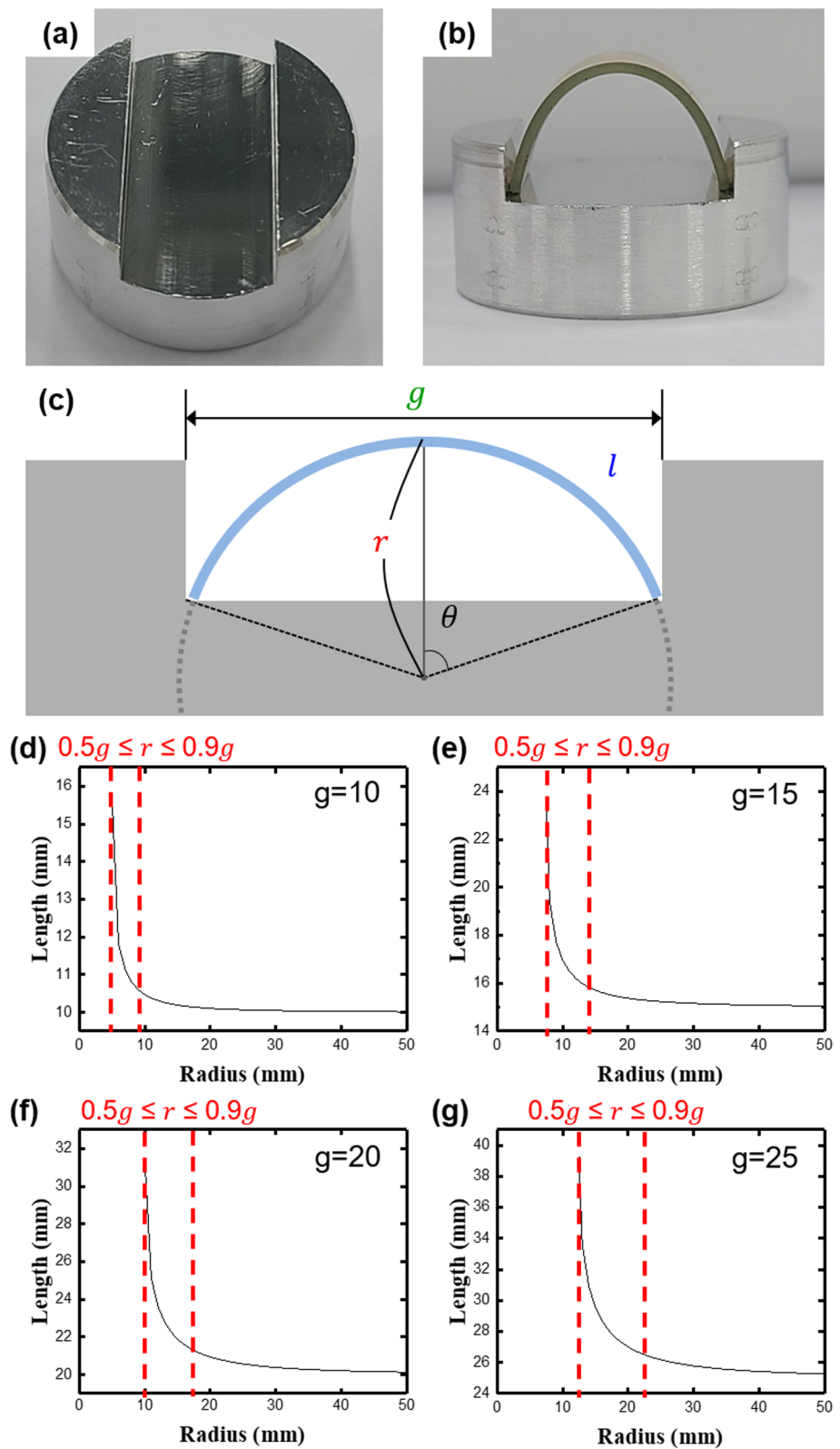
Direct observation of cracks under bending state requires that the flexible device be positioned within the SEM(Scanning Electron Microscope) while maintaining its bent form. Previous studies have employed large and complex equipment to control the bending radius of flexible devices [21–23]. The limited specimen space inside the SEM makes it challenging to use these existing devices. This study proposes a simple and compact jig suitable for use within the SEM, as shown in Fig. 1a. The jig is designed in the form of a cylinder with a groove and allows the flexible device to be inserted and retained in the desired bending state, as depicted in Fig. 1b. Based on the geometric relationship illustrated in Fig. 1c, the bending radius (r) can be determined by the gap of a jig (g) and the specimen length (l). Equation (1) was derived based on the arc length expression and a trigonometric function. This equation indicates that, with a fixed gap of a jig, the bending radius can be controlled by varying the specimen length.

$$l = \frac{\pi r \sin^{-1}\left(\frac{g}{2r}\right)}{90} \quad (1)$$

Though the bending radius increases as the specimen length decreases in Eq. (1), the change in specimen length becomes negligible to the extent when the bending radius exceeds about 90% of the gap of a jig as shown in Fig. 1d–g. For instance, when the gap of a jig is 10 mm, increasing the bending radius from 5 to 10 mm results in an approximately 33.33% reduction in specimen length, whereas increasing it from 10 to 20 mm leads to only a 3.48% decrease. This means that the maximum bending radius is about 90% of the gap of a jig. The bending radius can't be smaller than the half gap of a jig where the specimen becomes flat, which means the minimum bending radius is 50% of the gap of a jig. The limited applicable bending radius range for a single jig requires the use of multiple gap of a jigs to implement a wider range of bending conditions. In this study, gap of a jigs of 10, 15, and 20 mm were selected to conduct experiments across a broader set of bending radii.

Under the set bending conditions, the application of deformation causes the top surface of the specimen (outer bending deformation) to experience tensile stress and the bottom surface (inner bending deformation) to experience compressive stress. Stress applied to the thin film deposited on the substrate exceeds a critical threshold, initiating crack formation. Greater bending deformation leads to increases

Fig. 1 **a** A proposed jig to maintain bending conditions for observing the specimen inside the SEM., **b** An image of the actual specimen mounted on the proposed jig. **c** A schematic diagram of the geometric relationship of the bending radius and sizes of the proposed jig and the specimen. Bending radius-specimen length graph showing the limited range of bending radii for each gap of a jig. **d** $g=10$ mm, **e** $g=15$ mm, **f** $g=20$ mm, **g** $g=25$ mm



in both width and number of cracks. Crack propagation reduces the internal mechanical contact area, resulting in greater resistance being measured for wider cracks and

smaller resistance for narrower ones. Previous studies [2–8, 10, 11, 24–26] have reported an increase in resistance in the flat state after bending deformation, associated to the

Table 1 Experimental conditions for bending radius (r), specimen length (l), and gap of the jigs (g) for crack observation

Bending radius (r) (mm)	Length (l) (mm)	Gap (g) (mm)	Specimen width (mm)
5	15.7	10	15
7.5	23.5	15	15
10	31.4	20	15
12	23.5	20	15

number of cracks. The number of cracks obviously increases resistance, however, it is necessary to normalize the number by length or area to fairly compare the number of cracks of different specimen, converting it into crack density. Higher crack density restricts electron movement, suggesting a proportional relationship between crack density and resistance. Flexible devices are often used under bending state, so the correlation between cracks and resistance under bending state should be analyzed. Therefore, this study focuses on the analysis of width and density of cracks under bending state, which is expected to significantly affect resistance changes.

3 Experiments

We selected a 1 mm thick thermoplastic polyurethane (TPU) as a substrate of a flexible device, commonly used in the fabrication of flexible devices [27–29]. A 26 nm thick Au coating was applied to the top surface using the sputtering method. The previously described jig is used, and the bending radii have a range from 5 mm to 12 mm with about 2.5 mm space by varying the gap of a jig. The specimen width was set to 15 mm. Detailed experimental conditions are shown in Table 1.

The surface of each specimen was observed by SEM (Jeol Ltd.) for analyze crack parameters before, under and after bending states. The surface before bending state was observed for confirming that there was no crack and all cracks under bending state were generated by bending deformation. We selected eight areas ($6.4 \mu\text{m} \times 4.8 \mu\text{m}$) randomly on the central region of the specimens, where surface was clear and distortion-free, and the size of the area was set to contain only one or two cracks for measuring the crack width accurately. Crack width was measured twice at the center of each crack in each area, and totally eight crack widths were measured per a specimen, and the average value was calculated. Crack density was calculated by drawing eight random lines, each 100 μm in length, on an area ($128 \mu\text{m} \times 96 \mu\text{m}$) and dividing the number of cracks intersecting each line by the length of the line. The size of the area was determined to have enough cracks for comparing the crack density of all specimen to each other. We selected eight areas for measuring the crack density and the average value was calculated in each specimen.

Resistance measurement with a multimeter became difficult when the specimen was too short in a jig designed for crack observation. To address this issue, another measurement setup was customized, as shown in Fig. 2a. This setup included a dedicated grip section to secure the specimen during bending. Considering the length occupied by the grip section, an additional 20 mm was added to the original specimen length to ensure that the target bending radius could be accurately applied as shown in Fig. 2b. The bending radius for resistance measurement was set by adding 8 mm instead of 5 mm, which exceeded the measurable resistance range. We reset the bending radii 7.5, 8, 10, and 12 mm. Among them, 8 mm was selected to impose the most severe bending condition, since bending deformation tended to increase exponentially as the radius decreases. Specimen width of

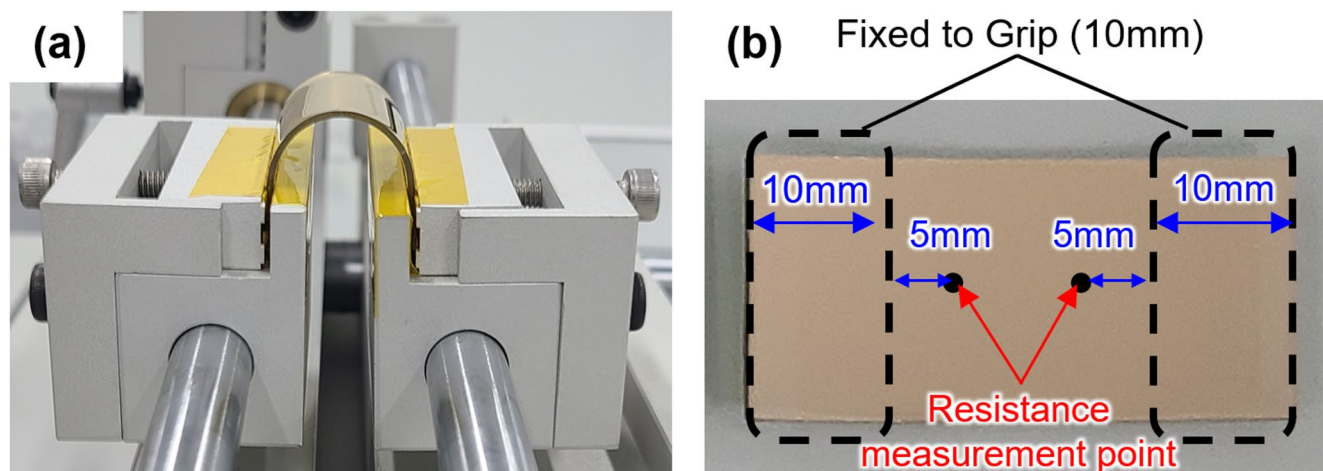


Fig. 2 **a** An image of the fixed specimen for resistance measurement under bending states, **b** A schematic diagram of the resistance measurement points

Table 2 Experimental conditions for bending radius, specimen length, and gap of the jigs for resistance measurement

Bending radius (r) (mm)	Length (l) (mm)	Gap (g) (mm)	Specimen width (mm)
7.5	43.5	15	30
8	45.1	16	30
10	51.4	20	30
12	43.5	20	30

30 mm was selected to provide sufficient contact area for reliable resistance measurement. Detailed specimen dimensions are summarized in Table 2. Resistance was measured at the center of the specimen where the multimeter probes made contact. Measurements were conducted at three stages

for each bending radius: before, under and after bending states.

4 Results and Discussion

SEM analysis revealed that no cracks were present before bending, as shown in Fig. 3, while cracks appeared under bending state across all bending radii. In general, when an external force is applied to a material, deformation progresses in the sequence of elastic deformation, plastic deformation, and fracture. The yield strength indicates the boundary between elastic and plastic deformation, while the tensile strength corresponds to the maximum stress a material can withstand within the plastic regime. In most

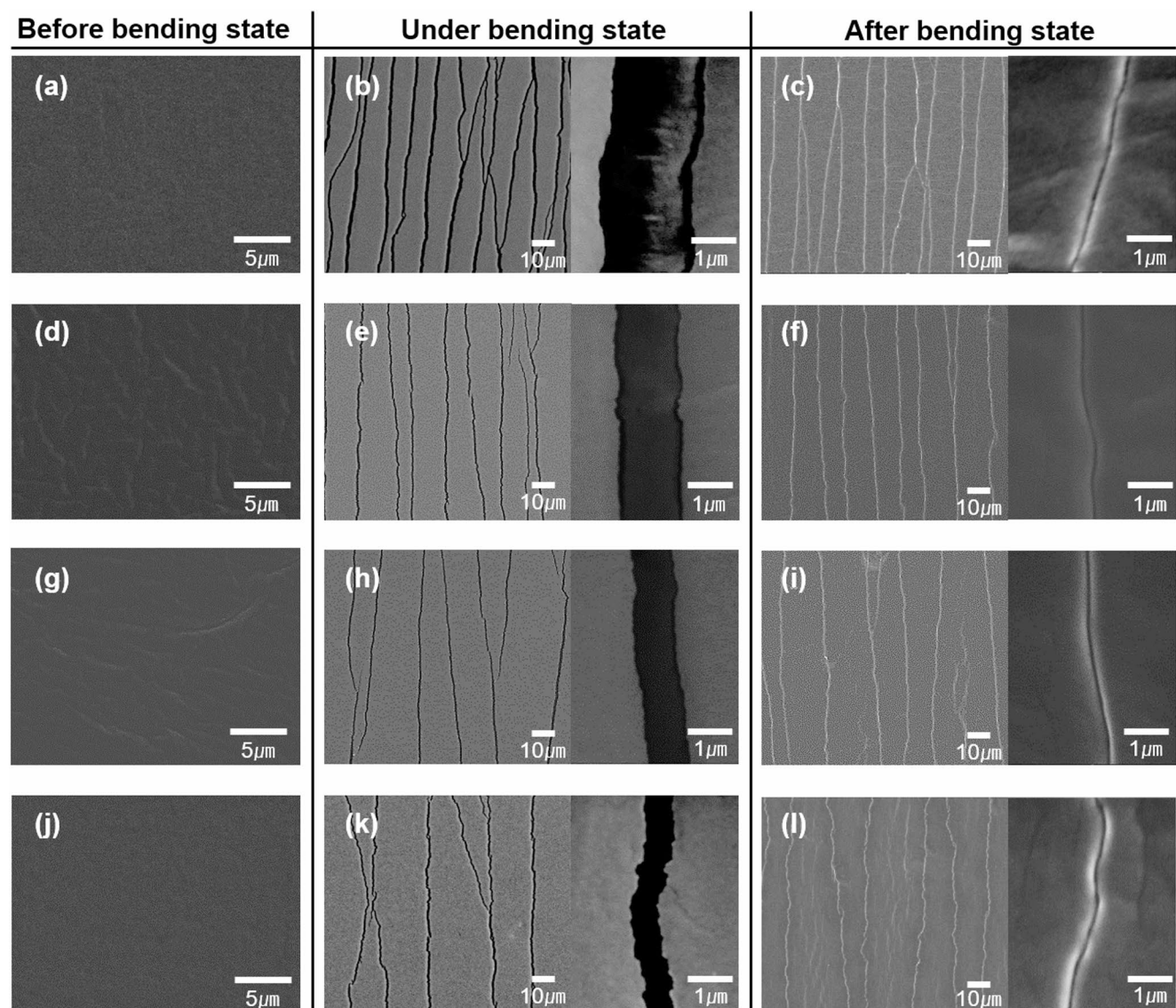


Fig. 3 SEM photos of crack density and crack width before, under and after bending states with varying bending radii. **a–c** $r=5$ mm **d–f** $r=7.5$ mm **g–i** $r=10$ mm **j–l** $r=12$ mm

conventional materials, plastic deformation initiates once applied stress exceeds the yield strength, and fracture occurs when it surpasses the tensile strength. According to previous studies [30], a 500 nm thick Au thin film exhibits a yield strength of 148 MPa and a tensile strength of 250 MPa. The yield strength for 200 nm thick films is 315 MPa, and the tensile strength is 383 MPa. These results demonstrate that as film thickness decreases, yield strength increases while the difference between yield and tensile strengths becomes smaller. This phenomenon is primarily attributed to the Hall-Petch effect, where thinner films exhibit smaller grain sizes that hinder dislocation motion and enhance strength. In addition to grain refinement, the increased influence of substrate constraint in thinner films further restricts dislocation mobility. As a result, plastic deformation is more limited, and fracture tends to occur shortly after the yield point. For example, in a 200 nm thick Au film, the difference between yield and tensile strengths is only 68 MPa. The 26 nm thick Au film used in this study is expected to show an even smaller difference, due to further enhancement in yield strength and a possible reduction in tensile strength at this scale. Based on this understanding, we assumed the yield strength could be similar to the threshold for the onset of fracture. The corresponding yield strain (ϵ_y) was calculated using Hooke's law, as shown in Eq. (2).

$$\epsilon_y = \frac{E}{\sigma_y} \quad (2)$$

Here, E represents the Young's modulus, and σ_y represents the yield strength. Based on the values presented in previous literature [30], we calculated the yield strain to be smaller than 0.01. Equation (3) defines the bending strain of a thin plate, where r is the bending radius and t is the specimen thickness.

$$\epsilon = \frac{t}{2r} \quad (3)$$

Table 3 Average values of crack density and crack width under bending state and after bending state

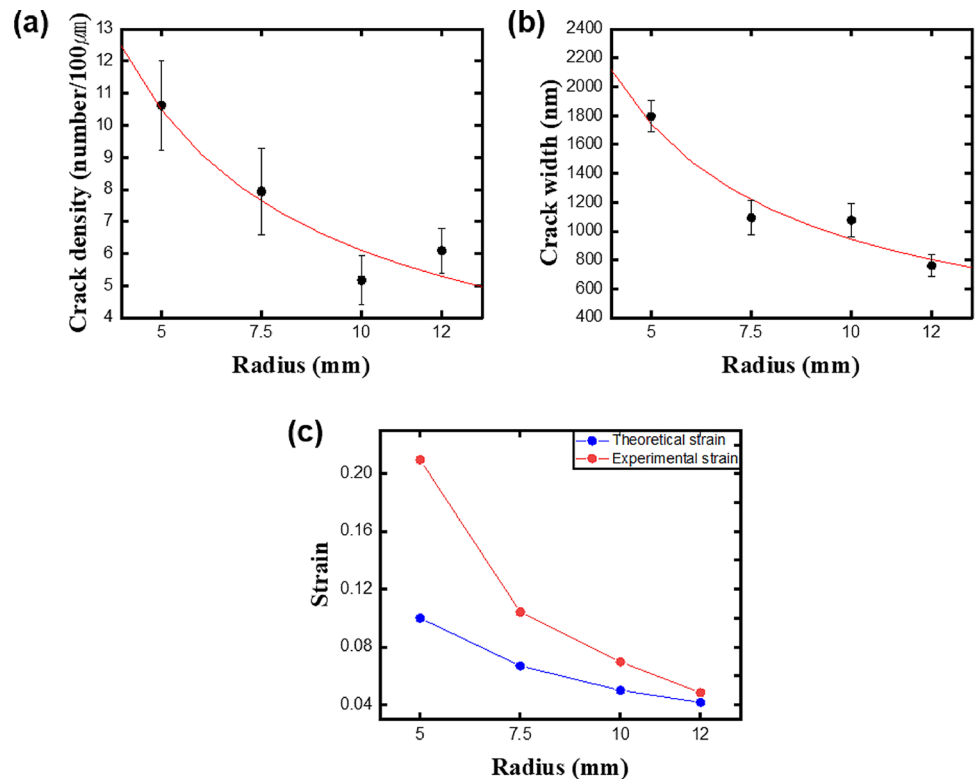
		Bending radius (mm)			
		5	7.5	10	12
Crack density (number/100 μ m)	Under bending state	10.63	7.94	5.17	6.09
	After bending state	10.25	8.70	5.66	6.25
Crack width (nm)	Under bending state	1795.2	1093.16	1076.26	761.37
	After bending state	58.55	59.46	53.70	55.10

Using this equation, the calculated strain values were 0.1, 0.067, 0.05, and 0.042 for bending radii of 5, 7.5, 10, and 12 mm, respectively. Comparison with the yield strain showed that all bending conditions induce strains exceeding the yield strain by at least a factor of four. Even though the yield strength was not the threshold for the onset of fracture, the factor of four was enough for the bending strain to exceed the threshold. The applied stress under bending state significantly surpasses the yield strength and the threshold, resulting in crack formation across all radii, which predominantly occurs on the outer bending side.

As shown in Fig. 3, crack density increased as the bending radius decreased. Bending radius of 5 mm showed an average of about 10 cracks per 100 μ m, while a radius of 12 mm resulted in about 6 cracks per 100 μ m, which was approximately a 40% reduction. Under the most severe deformation condition, with a bending radius of 5 mm, the crack densities measured 10.63 cracks per 100 μ m under bending state and 10.25 cracks per 100 μ m after bending state. This behavior was interpreted as evidence of irreversible damage. Figure 3 also showed that crack width, like crack density, increased as the bending radius decreased under bending state. The average crack width at 5 mm was approximately 1,800 nm, while at 12 mm it was about 760 nm, corresponding to a reduction of approximately 60%. However, after returning to the flat state, crack width nearly recovered and remained consistent across all bending radii, ranging from approximately 55 to 60 nm. This recovery behavior indicated that crack width is reversible damage. These observations suggested that a smaller bending radius increased strain, which in turn induced higher tensile stress on the specimen surface, and this stress increase resulted in greater crack density and wider crack widths. Notably, crack density exhibited irreversible behavior, while crack width showed reversible characteristics upon flattening. The measured values of crack density and crack width for each bending radius and deformation state are summarized in Table 3.

To quantify the relationship between crack characteristics and bending radius, we conducted regression analysis on the measured crack width and crack density. As shown in Fig. 4, both parameters exhibited non-linear trends with respect to bending radius. A logarithmic transformation was applied to the data to linearize the relationships, and correlation equations were subsequently derived. The analysis revealed that crack density scaled with the bending radius raised to the power of -0.78 , while crack width scaled with the bending radius raised to the power of -0.91 . These results indicate that crack width was a little more sensitive to changes in bending radius than crack density. Some deviations from the overall trend were observed. For example, in Fig. 4a, the crack density at a bending radius of 10 mm was lower than that at 12 mm. Similarly, in Fig. 4b, the crack width at

Fig. 4 **a** A graph of the correlation between crack density and bending radius, **b** A graph of the correlation between crack width and bending radius. **c** A graph of the comparison of theoretical and experimental strain from Eqs. (3) and (5), respectively



10 mm was comparable to that at 7.5 mm. These differences may be attributed to the interaction between crack width and crack density. When cracks formed due to bending deformation, the tensile stress on the material surface was partially relieved through localized stretching around the cracks. Both crack width and crack density contributed to this stress relief. A Larger crack width reduced the number of cracks required to relieve the stress, resulting in a lower crack density. Conversely, a smaller crack width required a greater number of cracks to relieve the same amount of stress, leading to a higher crack density. This inverse relationship supported the hypothesis that crack width and crack density were correlated. Assuming that the total opening displacement of all cracks formed during bending was equal to the surface elongation of the specimen in Eq. (4), a mathematical relationship between crack width and crack density was established as Eq. (5), where l_0 represents the initial length of the specimen.

$$\begin{aligned}\Delta l &= \text{Crack width} \times \text{Number of crack} \\ &= \text{Crack width} \times \text{Crack density} \times l_0\end{aligned}\quad (4)$$

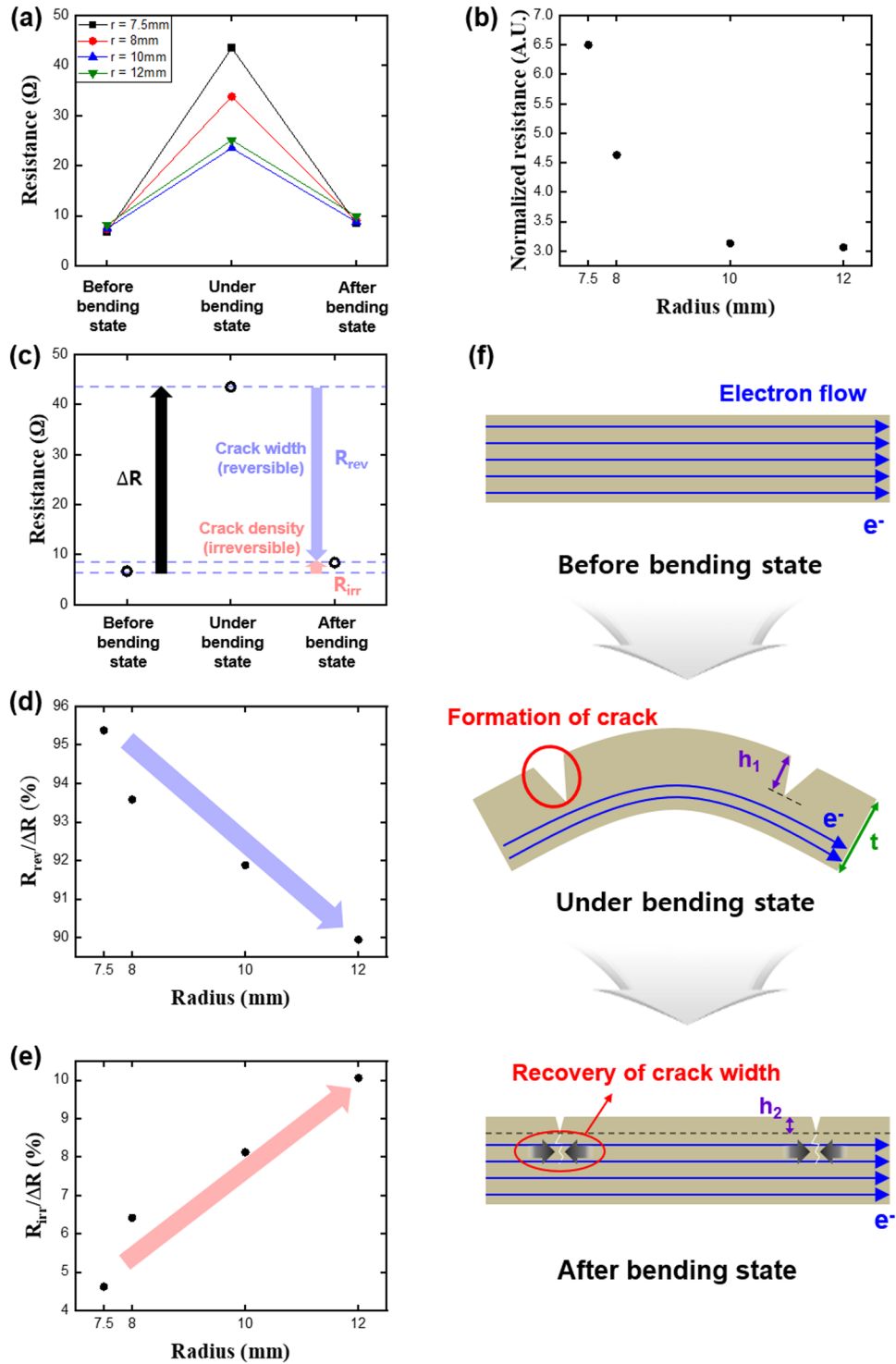
$$\frac{\Delta l}{l_0} = \varepsilon = \text{Crack width} \times \text{Crack density} \quad (5)$$

The experimentally calculated strain from Eq. (5) was compared with theoretical predictions based on the bending theory of a plate from Eq. (3) in Fig. 4c. As the bending radius decreased, both values increased. However, the

experimental values rose more quickly, resulting in a widening gap between theory and experiment. This discrepancy can be explained by the mechanical constraints acting on the cracks. In the early stages of crack growth, when the cracks had not yet penetrated the substrate, they were constrained by both the thin film and the underlying substrate. As the bending radius decreased and the strain increased, the cracks propagated deeper and eventually reached the substrate, removing the constraint from the thin film. This allowed the cracks to widen more significantly, thereby increasing the measured crack width at a faster rate. This behavior was consistent with the trend observed in Fig. 4b and c. Additionally, Eq. (4) assumes that all cracks are oriented perfectly perpendicular to the bending direction. In practice, however, cracks were often slightly tilted. As a result, more cracks are needed to relieve the applied stress than the theoretically predicted values. These findings indicate that at smaller bending radii, both crack width and crack density increase beyond theoretical expectations. This deviation is attributed to reduced mechanical constraints and geometric deviations in crack orientation.

We quantitatively analyzed how crack parameters affected electrical properties by measuring resistance under various bending conditions, as shown in Fig. 5a, b. At a bending radius of 5 mm, the resistance under bending state exceeded the measurable range and was therefore excluded from the measurement. Instead, a radius of 8 mm was included to enable meaningful comparison. Figure 5a

Fig. 5 **a** Resistance changes before, under and after bending states for various bending radius conditions. **b** Resistance increase rate for each bending radius under bending states. **c** The influence of crack parameters on total resistance change at a bending radius of 7.5 mm. **d** Contribution of resistance change as a function of bending radius with crack width, **e** Contribution of resistance change as a function of bending radius with crack density. **f** Schematic diagrams showing the relationship between crack depth, conductive layer thickness, and the amount of recovery



showed the resistance changes before, under and after bending states. As the bending radius decreased, resistance under bending state increased more significantly, while the resistance of after bending state remained relatively stable at all bending radii. Initial resistance varied depending on bending radius, thus we calculated the normalized resistance increase for fair comparison, as shown in Fig. 5b. This

trend was consistent with earlier crack observations, where smaller bending radii led to increased crack width and density, resulting in a larger resistance increase.

To further analyze the contributions of each crack parameter, we divided the total resistance change into two components (reversible and irreversible) and quantified them as shown in Fig. 5c. The recovered resistance between

under and after bending states was defined as R_{rev} , and the unrecovered portion as R_{irr} . Their respective contributions were calculated as ratios, $R_{rev}/\Delta R$ and $R_{irr}/\Delta R$, as shown in Fig. 5d and e. This reversible component R_{rev} is primarily influenced by crack width, as previously discussed. In contrast, R_{irr} reflected the unrecovered resistance and was associated with crack density. As shown in Fig. 5d and e, the contribution from crack width was consistently greater than that from crack density across all bending radii. At a bending radius of 7.5 mm, crack width accounted for about 95% of the total resistance change, while crack density contributed only about 5%. As bending radius increased, the contribution of crack width slightly decreased, however, even at 12 mm, it remained dominant at approximately 90%. These results indicated that the resistance increase in the flexible specimen under bending state in this study was mainly governed by crack width, rather than crack density.

Previous studies have mainly focused on analyzing cracks in the flat state (after bending state), often neglecting the critical influence of crack width on electrical resistance under bending state. Our results showed that crack width increased resistance variation much more than crack density did. Furthermore, the extent of crack width recovery had a significant effect on changes in electrical properties under and after bending states. The recovery of crack width induces mechanical contact in the thickness direction, which consequently leads to a decrease in crack depth. To quantify this recovery behavior of crack width, we introduced a new parameter based on the ratio of crack depth to total film thickness in Eq. (6) and visually illustrated in the schematic diagram in Fig. 5f.

$$\text{Amount of recovery} = \frac{h_1 - h_2}{t} \quad (6)$$

The variable t represents the total thickness of the thin film, h_1 denotes the crack depth under bending state, and h_2 indicates the crack depth after bending state. Two key factors contribute to increased crack width recovery. First, greater strain leads to a larger change in crack depth. As the bending radius decreases and strain increases, both crack width and crack depth grow, resulting in enhanced recovery. Second, a thinner film increases the ratio of crack depth to total thickness, which makes the recovery effect more pronounced. In summary, smaller bending radii and thinner conductive layers amplify the effects of crack width recovery. Smaller bending radii and thinner layers are consistent with current trends in flexible displays, where high flexibility and reduced thickness are essential. Therefore, accounting for crack width behavior is increasingly important in the design of such devices.

5 Conclusions

This study analyzed crack parameters and electrical property changes before, under and after bending states similar to those experienced by flexible devices. Au with 26 nm thickness was deposited on a 1 mm TPU substrate to apply bending deformation, and analysis under bending state from 5 mm to 10 mm bending radii led to the following conclusions:

- (1) Crack formation at all tested bending radii was analyzed based on yield strength assuming that it was similar to the threshold for the onset of fracture. Yield strain was calculated using previously reported values and compared with the applied strain. In all cases, the applied strain exceeded the yield strain by at least a factor of four, indicating that applied stress at all bending radii could occur fracture and formed cracks.
- (2) As the bending radius decreased, tensile stress on the specimen surface increased, leading to growth in both crack width and crack density. Crack width showed substantial recovery indicating reversible deformation while crack density remained unchanged, representing irreversible damage after bending state. Among the two parameters, crack width showed a little greater sensitivity to changes in bending radius.
- (3) Both crack width and crack density showed negative exponential relationships with the bending radius, and crack width was a little more sensitive to changes in the bending radius. During crack formation, the two variables influence each other, and a mathematical relationship can be derived under the assumption that the sum of all crack widths equals the tensile strain. In practice, cracks are not perfectly vertical, necessitating the formation of additional cracks to relieve stress. As a result, at smaller curvature radii, both crack width and density are observed to be greater than theoretical predictions.
- (4) Resistance measurements at various bending radii revealed that resistance increased as the bending radius decreased. The resistance increased under the bending state and then significantly decreased after the bending state due to the recovery of crack width. Therefore, if the resistance increase is measured after bending state, the increase may be underestimated compared to that under bending state, which corresponds to the actual in-use condition, thereby potentially leading to an overestimation of the reliability. Such underestimation of the resistance increase becomes more pronounced when the bending radius is smaller or the conductive layer is thinner.

Author Contributions Dong-Hyeon Kim: Conceptualization, Investigation, Data analysis, Visualization, Writing-original draft. Ji-Youn

Kwak: Methodology, Writing-review & editing. Ju-Young Kim: Supervision, Funding acquisition., Methodology. Eun-chaee Jeon: Supervision, Funding acquisition, Conceptualization, Methodology, Validation, Writing-review & editing.

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Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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